BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DG KHAN

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 8193 / 8196

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

OBJECTIVE

Time Allowed: 30 MINUTES | **Maximum Marks:** 20 | **Paper Code:** 8193

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct , fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. Derivative of √x w.r.t. x at x = a is

(A) √a
(B) 2√a
(C) 1/√a
(D) 1/(2√a)

Q2. If $f(x) = x^{100}$, f'(1) =

(A) 0
(B) 50
(C) 99
(D) 100

Q3. ∫ a^λx dx =

(A) a^λx/λ + c
(B) a^λx/ln a + c
(C) a^λx/(λ ln a) + c
(D) a^λx . ln a + c

Q4. $\int e^x (1/x - 1/x^2)\, dx =$

(A) $e^x/x + c$
(B) $- e^x/x + c$
(C) e^x. ln x + c
(D) $- e^x/x^2 + c$

Q5. $\int 1/(x^2 + 16)\, dx =$

(A) $\tan^{-1}(x/4) + c$
(B) $1/4 \tan^{-1}(x/4) + c$
(C) $1/4 \tan(x/4) + c$
(D) $1/2 \tan^{-1}(x/4) + c$

Q6. $\int 0\, dx =$

(A) 0
(B) 1
(C) x + c
(D) constant

Q7. A line which pass through one vertex and mid-point of opposite side of a triangle is called

(A) Median
(B) Altitude
(C) Normal
(D) Perpendicular bisector

Q8. If A(-2 , 3) , B(-4 , 1) and C(3 , 5) are the vertices of a triangle , then its centroied is

(A) (-3/2 , 9/2)
(B) (-1 , 3)
(C) (-3 , 4)
(D) (-3 , 9)

Q9. If point (2 , -9) lies on line px + y + 20 = 0 , then value of p is

(A) 11/2
(B) -11/2
(C) 29/2
(D) -29/2

Q10. If x > b , then which one is correct ?

(A) -x > -b
(B) -x < b
(C) x < b
(D) -x < -b

Q11. The circle whose radius is 0 is called a/an

(A) Unit circle
(B) Imaginary circle
(C) Point circle
(D) Circum circle

Q12. The point (-5 , 6) lies ..... the circle $x^2 + y^2 = 61$

(A) Outside
(B) Inside
(C) On
(D) Any where

Q13. The length of semi-latus rectum of hyperbola

(A) 2a
(B) $b^2/2a$
(C) b^2/a
(D) $2b^2/a$

Q14. Which of the following is not vector quantity

(A) Weight
(B) Momentum
(C) Force
(D) Energy

Q15. If vectors a and b have same direction , then a . b =

(A) ab
(B) -ab
(C) ab sinθ
(D) $(ab)^2$

Q16. Value of $2i \times 2j \cdot k$ is

(A) 0 (B) 1

(C) 2 (D) 4

Q17. cosech hx is equal to .....

(A) $2/(e^x + e^{-x})$ (B) $1/(e^x + e^{-x})$

(C) $2/(e^x - e^{-x})$ (D) $2/(e^{-x} - e^x)$

Q18. $f(x) = ax + b$, $a \neq 0$ is a/an

(A) Linear function (B) Odd function

(C) Even function (D) Identity function

Q19. Derivative of an identity function is

(A) 0 (B) 1

(C) -1 (D) Identity function

Q20. x^3 d/dx (ln 2x) =

(A) x^2 (B) $2x^3$

(C) $3x^2$ (D) $6x^2$

SUBJECTIVE SECTION-I / SECTION-II

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Express perimeter 'p' of a square as a function of its area 'A'

ii Without finding inverse state domain and range of f^{-1} if $f(x) = (x - 5)^2$, $x \geq 5$

iii Evaluate $\lim_{x \to 1} (x^2 - 1)/(x^2 - x)$

iv Evaluate the limit $\lim_{\theta \to 0} \sin^2\theta/\theta$

v Differentiate with respect to 'x' $1/(x - a)$ by definition

vi Differentiate with respect to 'x' $(a + x)/(a - x)$

vii Find dy/dx by making suitable substitution of $y = (3x^2 - 2x + 7)^6$

viii Differentiate with respect to 'x' $1/a \sin^{-1}(a/x)$

ix Differentiate $(\ln x)^x$ with respect to 'x'

x Find y_2 if $x^2 + y^2 = a^2$

xi Show that $\cos(x+h) = \cos x - h \sin x - h^2/2! \cos x + h^3/3! \sin x + ...$

xii Find interval in which ' f ' is increasing or decreasing $f(x) = \cos x$, $x \in (-\pi/2 , \pi/2)$

Q3. (Marks 16) Write short answers any Eight (8) of the following

i Find δy and dy of $y = x^2 - 1$, when x changes from 3 to 3.02

ii Evaluate $\int (\sqrt{\theta} - 1)^2/\sqrt{\theta}\, d\theta$

iii Find the area between the x-axis and the curve $y = 4x - x^2$

iv Solve the differential equation $dy/dx = y/x^2$, $(y > 0)$.

v Evaluate $\int_{-1}^{3} (x^3 + 3x^2)\, dx$

vi Evaluate $\int x \ln x\, dx$

vii Find $\int (-2x)/\sqrt{(4 - x^2)}\, dx$

viii Find distance between the points A(-8 , 3) ; B(2 , -1). Also find mid-point between them

ix The coordinates of a point p are (-6 , 9). The axes are translated through the point O' (-3 , 2). Find the coordinates of P referred to the new axes

x Show that points (-4 , 6) , (3 , 8) and (10 , 10) lie on the same line

xi Find the distance from the point P(6 , -1) to the line $6x - 4y + 9 = 0$

xii Find measure of the angle between the lines represented by $x^2 - xy - 6y^2 = 0$

Q4. (Marks 18) Write short answers any Nine (9) of the following

i Graph the inequality $x + 3y > 6$

ii Define feasible region and feasible solution

iii Find the centre and radius of circle $x^2 + y^2 - 6x + 4y + 13 = 0$

iv Find the slope of normal to the circle $x^2 + y^2 = 25$ at (4 , 3)

v Check the position of the point (5 , 6) w.r.t circle $x^2 + y^2 = 81$

vi Find the focus and directrix of parabola $x^2 = -16y$

vii Find centre and foci of ellipse $25x^2 + 9y^2 = 225$

viii Find eccentricity and vertices of $x^2 - y^2 = 9$

ix Find a vector whose magnitude is 2 and is parallel to $-i + j + k$

x Find cosine of the angle between u and v where u = [-3 , 5] and v = [6 , -2]

xi Compute $a \times b$ and $b \times a$ if $a = i + j$ and $b = i - j$

xii If $a + b + c = 0$ then prove that $a \times b = b \times c$

xiii Find the volume of the parallelepiped determined by $u = i + 2j - k$, $v = i - 2j + 3k$ and $w = i - 7j - 4k$

> Note: SECTION-II — Note: Attempt any Three questions from this section 10 x 3 = 30

Q5.

(A) Discuss continuity of f at x = 3 , when f(x) = { x - 1 if x < 3 ; 2x + 1 if x ≥ 3 }

(B) Prove that y dy/dx + x = 0 if x = (1 - t^2)/(1 + t^2) , y = 2t/(1 + t^2)

Q6.

(A) If y = $(\cos^{-1}x)^2$, prove that (1 - x^2) y_2 - x y_1 - 2 = 0

(B) Evaluate : ∫ √(4 - $5x^2$) dx

Q7.

(A) Evaluate ∫(0 to π/4) (cosθ + sinθ)/($2\cos^2\theta$) dθ

(B) Maximize f(x , y) = x + 3y subject to the constraints 2x + 5y ≤ 30 ; 5x + 4y ≤ 20 ; x ≥ 0 ; y ≥ 0

Q8.

(A) Find equations of the tangents drawn from (0 , 5) to $x^2 + y^2 = 16$

(B) Prove that sin(α - β) = sin α cos β - cos α sin β using vectors

Q9.

(A) Find centre , foci , eccentricity and directrices of hyperbola $x^2/16 - y^2/9 = 1$

(B) Find equation of line through the intersection of x - y - 4 = 0 and 7x + y + 20 = 0 and perpendicular to the line 6x + y - 14 = 0

GROUP – II

OBJECTIVE

Time Allowed: 30 MINUTES	**Maximum Marks:** 20	**Paper Code:** 8196

Note: You have four choices for each objective type question as A , B , C and D . The choice which you think is correct , fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question.

Q1. ∫ e^x(sinx − cosx) dx = ?

(A) e^x cosx + c
(B) e^x sinx + c
(C) − e^x cosx + c
(D) − e^x sinx + c

Q2. ∫(0 to 1/2) 1/√(1−x^2) dx = ?

(A) π/6
(B) π/4
(C) π/3
(D) π/2

Q3. The distance of a point P(2 , -3) from the x-axis is equal to

(A) -3
(B) -2
(C) 2
(D) 3

Q4. If (2 , 4) , (4 , 6) and (3 , 2) are the vertices of a triangle , then coordinates of the centroid are

(A) (3 , 4)
(B) (4 , 6)
(C) (9/2 , 6)
(D) (24 , 48)

Q5. The lines represented by $3x^2 - 5xy - 3y^2 = 0$ will be

(A) Parallel
(B) Perpendicular
(C) Neither parallel nor perpendicular
(D) Tangent lines

Q6. x = 2 is the solution of

(A) $x > 1$
(B) $x < 5$
(C) $x > 7$
(D) $x > 9$

Q7. A chord which contains the centre of the circle is called

(A) Radius
(B) Focal chord
(C) Diameter
(D) Tangent line

Q8. The perpendicular at the outer end of a radial segment is ..... to the circle

(A) Secant
(B) Normal
(C) Perpendicular
(D) Tangent

Q9. Asymptotes of the curve $x^2/16 - y^2/25 = 1$ are

(A) $y = \pm 5/4\ x$
(B) $y = \pm 4/5\ x$
(C) $y = \pm 5/4\ \sqrt{(x^2 - 16)}$
(D) $y = -5/4\ \sqrt{(x^2 - 16)}$

Q10. Projection of a vector b along vector a is

(A) (a.b)/|b|
(B) (a.b)/b
(C) (a.b)/a
(D) (a.b)/|a|

Q11. The value of [k i j] = ?

(A) -1
(B) 0
(C) 1
(D) 2

Q12. If three vectors a , b and c are coplanar , then scalar triple product of these vectors is

(A) a negative number
(B) a positive number
(C) a non-negative number
(D) zero

Q13. Lim (x → a) (x^n-1 − a^n-1)/(x − a) = ?

(A) na^n-1 (B) (n-1)a^n-2

(C) na^n-1 (D) (n-1)a^n-1

Q14. If f(x) = 2 + √(x − 1) ∀ x ∈ IR then domain of $f^{-1}(x)$ is

(A) [-1 , +∞) (B) [0 , +∞)

(C) [1 , +∞) (D) [2 , +∞)

Q15. d/dx (x − sin2x/2) = ?

(A) $2\sin^2 x$ (B) $2\cos^2 x$

(C) 2sinx (D) 2cosx

Q16. If f(x) = 1/12 x^4 , then $f^{(4)}(x)$ = ?

(A) 0 (B) 1

(C) 2 (D) 3

Q17. If xy + y^2 = 2 , then dy/dx = ?

(A) (−x)/(x + 2y) (B) (−y)/(x + 2y)

(C) (xy − y)/(x + 2y) (D) (x − 2y)/(x − y)

Q18. If f(x) = x^2 + 2x − 3 , then f(x) is decreasing in the interval

(A) (-1 , +∞) (B) (-∞ , -1)

(C) (-∞ , 1) (D) (1 , 3)

Q19. ∫ (sinx − cosx)/√(1 − sin2x) dx = ?

(A) x + c (B) sinx + c

(C) cosx + c (D) $\cos^2 x$ + c

Q20. ∫ x/(x + 2) dx = ?

(A) x + ln(x + 2) + c (B) x − $\ln(x + 2)^2$ + c

(C) x − ln(x + 2) + c (D) x + $\ln(x + 2)^2$ + c

SUBJECTIVE SECTION-I / SECTION-II

Q2. (Marks 16) Write short answers any Eight (8) of the following

i Express the perimeter P of a square as a function of its area A

ii Find the values of (fog) and (gof) when f(x) = 2x + 1 , g(x) = 3/(x - 1)

iii Evaluate lim (x → -1) (x^3 - x)/(x + 1)

iv Find c such that lim (x → -1) f(x) exist where f(x) = { x + 2 , x ≤ -1 ; c + 2 , x > -1 }

v Find dy/dx by definition when y = 2x^2 + 1

vi Find dy/dx when y = (2x - 3)/(2x + 1)

vii If x = θ + 1/θ and y = θ + 1 , find dy/dx

viii Differentiate sin x w.r.t. cos x

ix If y = x e^(sin x) , find dy/dx

x Find y_2 when $x = at^2$, $y = bt^4$

xi Find the extreme values of $f(x) = 3x^2$

xii Find y_2 when $y = 2x^5 - 3x^4 + 4x^3 + x - 2$

Q3. (Marks 16) Write short answers any Eight (8) of the following

i Use differentials to find dy/dx and dx/dy of $x^4 + y^2 = xy^2$

ii Evaluate $\int (1 - \sqrt{x})^2/\sqrt{x}\, dx$

iii Evaluate $\int dx/(x^2 + 4x + 13)$

iv Evaluate $\int x^2 \tan^{-1} x\, dx$

v Evaluate $\int ((a-b)x)/((x-a)(x-b))\, dx$

vi Evaluate $\int_1^2 (x^2 + 1)/(x + 1)\, dx$

vii Solve the differential equation dy/dx = (1 - x)/y

viii Show that points A(-1 , 2) , B(7 , 5) and C(2 , -6) are vertices of right triangle

ix In a triangle A(8 , 6) , B(-4 , 2) , C(-2 , -6) find slope of any one median of triangle

x Find the slopes of lines l_1 and l_2 where l_1 : Joining (2 , 7) and (7 , 10) ; l_2 : Joining (1 , 1) and (-5 , 3)

xi Find the lines represented by $3x^2 + 7xy + 2y^2 = 0$

xii Find the distance between parallel lines 2x + y + 2 = 0 , 6x + 3y - 8 = 0

Q4. (Marks 18) Write short answers any Nine (9) of the following

i Indicate the solution set of the system of linear inequalities $3x + 7y \geq 21$, $x - y \leq 2$

ii Define feasible region

iii Find centre and radius of the circle $4x^2 + 4y^2 - 8x + 12y - 25 = 0$

iv Find vertex and directrix of parabola $(x - 1)^2 = 8(y + 2)$

v Define axis of parabola

vi Find an equation of hyperbola with foci (0 , ±6) and e = 2

vii Find centre and vertices of ellipse $25x^2 + 9y^2 = 225$

viii Find equation of tangent to the conic $y^2 = 4ax$ at point (x_1 , y_1)

ix Find direction cosines of the vector $6i - 2j + k$

x If the vectors $u = \alpha i + 2\alpha j - k$ and $v = i + \alpha j + 3k$ are perpendicular. Find the value of α

xi Define unit vector. Also give an example

xii Find the value of α for which $\alpha i + j$, $i + j + 3k$ and $2i + j - 2k$ are coplanar

xiii Define cross product of two vectors u and v

> Note: SECTION-II — Note: Attempt any Three questions from this section 10 x 3 = 30

Q5.

(A) Prove that $\lim (x \to 0) (a^x - 1)/x = \log_e a$

(B) If $x = a \cos^3\theta$, $y = b \sin^3\theta$, show that $a\, dy/dx + b \tan \theta = 0$

Q6.

(A) If $y = e^x \sin x$, show that $d^2y/dx^2 - 2\, dy/dx + 2y = 0$

(B) Evaluate : $\int \sqrt{(a^2 + x^2)}\, dx$

Q7.

(A) Evaluate $\int(0 \text{ to } 1) (3x)/\sqrt{(4 - 3x)}\, dx$

(B) Maximize $f(x , y) = x + 3y$ subject to constraints $2x + 5y \leq 30$; $5x + 4y \leq 20$; $x , y \geq 0$

Q8.

(A) Show that the circles $x^2 + y^2 + 2x - 2y - 7 = 0$ and $x^2 + y^2 - 6x + 4y + 9 = 0$ touches externally

(B) Use vector method to prove that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

Q9.

(A) Find an equation of the ellipse with given data centre (0 , 0) , focus (0 , -3) , vertex (0 , 4)

(B) If two vertices of an equilateral triangle are A(-3 , 0) and B(3 , 0). Find the third vertex. How many of these triangles are possible ?